

Fake Co-visitation Injection Attacks to Recommender Systems

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Abstract:

Recommender systems have become an essential component in a wide range of web services. It is believed that recommender systems recommend a user items (e.g., videos on YouTube, products on Amazon) that match the users preference. In this work, we propose new attacks to recommender systems. Our attacks exploit fundamental vulnerabilities of recommender systems and can spoof a recommender system to make recommendations as an attacker desires. Our key idea is to inject fake co-visitations to the system. Given a bounded number of fake co-visitations that an attacker can inject, two key challenges are 1) which items the attacker should inject fake co-visitations to, and 2) how many fake co-visitations an attacker should inject to each item. We address these challenges via modelling our attacks as constrained linear optimization problems, by solving which the attacker can perform attacks with maximal threats. We demonstrate the feasibility and effectiveness of our attacks via evaluations on both synthetic data and real-world

recommender systems on several popular web services including YouTube, eBay, Amazon, Yelp, and LinkedIn. We also discuss strategies to mitigate our attacks.

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